$$\text{binomial therom: } (x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

let $a=1$, 1^n always =1 so all x terms are just multiplied by 1

$$(x+1)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b 1\right]$$

let $x' = x-1$ to cancel the $+1$ and produce x^r

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}\left[\binom{r}{b}(x-1)^b\right]$$

bring out the $(x-1)^r$ to get recurssion (by applying the identity to that term)

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-1)^b\right] + (x-1)^r$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-1)^b\right] + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2)^b\right] + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-3)^b\right] + (x-3)^r$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-1)^b\right] + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2)^b\right] + ... + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-x)^b\right] + (x-x)^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

apply the identity to the b^c terms assymetrically to get sums of x^0

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{0}b^0 + \binom{r}{1}b^1 + \binom{r}{2}b^2 + ... + \binom{r}{r-1}b^{r-1}\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}b^0 + \binom{r}{1}\sum_{c=0}^{b-1}\left[\binom{1}{0}c^0\right] + \binom{r}{2}\sum_{c=0}^{b-1}\left[\binom{2}{0}c^0 + \binom{2}{1}\sum_{d=0}^{c-1}\left[\binom{1}{0}d^0\right]\right] + ... + \binom{r}{r-1}b^{r-1}\right] + 0^r$$

$$x^r = \binom{r}{0}\sum_{b=0}^{x-1}[b^0] + \binom{r}{1}\binom{1}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + \binom{r}{2}\binom{2}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + \binom{r}{2}\binom{2}{1}\binom{1}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[d^0] + ... + \binom{r}{r-1}b^{r-1}] + 0^r$$

$$x^r = \binom{r}{0}\sum_{b=0}^{x-1}[b^0] + \left(\binom{r}{1}\binom{1}{0} + \binom{r}{2}\binom{2}{0}\right)\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + \binom{r}{2}\binom{2}{1}\binom{1}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[d^0] + ... + \binom{r}{r-1}b^{r-1}] + 0^r$$

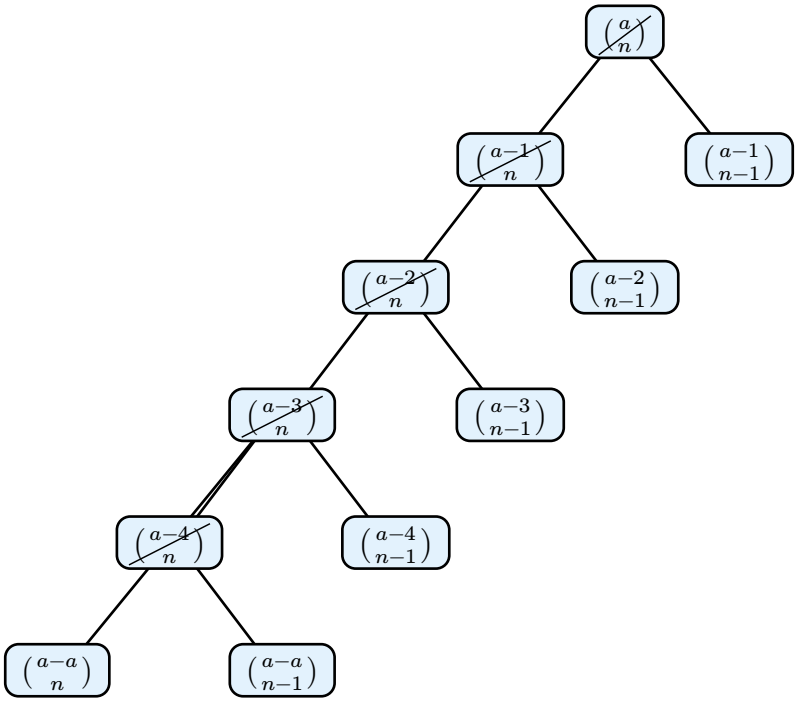
let the product and sum of these outer binom terms grouped by sum level (n), $= D_{r,n}$

$$x^r = D_{r,1}\sum_{b=0}^{x-1}[b^0] + D_{r,2}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + D_{r,3}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[d^0] + ... + D_{r,r}\overbrace{\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}...\sum_{n_n=0}^{n_{n-1}-1}[n_n^0]}^{r\text{ sums}} + 0^r$$

pascals identity

$$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$$

apply pascals identity recursivley to lhs



$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\binom{b}{n-1}\right] + \binom{0}{n}$$

apply the identity to the inner $\binom{b}{n-1}$

$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\binom{c}{n-2}\right] + \binom{0}{n-1}\right] + \binom{0}{n}$$

expanding the inner term n-1 times, for n total sums, causes its inner term to be 1, as $\binom{a \geq 0}{0} = 1$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\dots \sum_{n_n=0}^{n_{n-1}-1}\left[\binom{n_n}{n-n}\right] + \binom{0}{n-(n-1)}\dots\right]\binom{0}{n-2}\right] + \binom{0}{n-1}\right]}_{\text{n sums}} + \binom{0}{n}$$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\dots \sum_{n_n=0}^{n_{n-1}-1}\left[\binom{n_n}{0}\right] + \binom{0}{n-(n-1)}\dots\right]\binom{0}{n-2}\right] + \binom{0}{n-1}\right]}_{\text{n sums}} + \binom{0}{n}$$

as we only expanded n-1 times the $\binom{0}{n_{n-1}}$ only reach $\binom{0}{n-(n-1)} = \binom{0}{1} = 0$, as $\binom{0}{n>0} = 0$

if $n = 0$, the sums arent expanded and only the $\binom{0}{n}$ remains

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\dots \sum_{n_n=0}^{n_{n-1}-1}[1]\right]\right]\right]}_{\text{n sums}} + \binom{0}{n}$$

the sums in the binomial therom expansion and pascals identity expansions have the same bounds change

$$x^r = D_{r,1}\sum_{b=0}^{x-1}[1] + D_{r,2}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[1] + D_{r,3}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[1] + ... + D_{r,r}\overbrace{\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}...\sum_{n_n=0}^{n_{n-1}-1}}^{r\text{ sums}}[1] + 0^r$$

$$x^r = D_{r,1}\binom{x}{1} + D_{r,2}\binom{x}{2} + D_{r,3}\binom{x}{3} + ... + D_{r,r}\overbrace{\binom{x}{r}}^{r\text{ sums}} + 0^r$$

$$x^r = \sum_{b=1}^{r}\left[D_{r,b}\binom{x}{b}\right] + 0^r$$

$$\text{let } D_{r,0} = 0^r$$

$$x^r = \sum_{b=0}^{r}\left[D_{r,b}\binom{x}{b}\right]$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

every expansion of b^c adds a sum layer to the expanded terms, the $\sum_{b=0}^{x-1}$ is this accumulator

each new layer of b_n^c adds a sum to the expanded terms, so each new layer can only produce diffrences upto n (beacuse it has a term with a minimum of n sums)

therfore to get the nth diffrence you would on need to expand n times, (n-1 times if you count the base identity as the 0th expansion beacuse it adds a outer sum to the terms); diliberate expansion of terms with r

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}b^0 + \binom{r}{1}b^1 + \binom{r}{2}b^2...\binom{r}{r-1}b^{r-1}\right] + 0^r$$

$$b^0 = 1$$

$$b^1 = \sum_{c=0}^{x-1}\left[\overbrace{\binom{1}{0}c^0}^{1}\right]$$

$$b^2 = \sum_{c=0}^{x-1}\left[\overbrace{\binom{2}{0}c^0}^{1} + \binom{2}{1}\sum_{d=0}^{c-1}\left[\overbrace{\binom{1}{0}d^0}^{1}\right]\right]$$

each layer contains a 1 term it will have L (layer) amount of sums (for $r \geq 0$)

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=1}^{r-1}\left[\binom{r}{c}b^c\right] + \overbrace{\overbrace{\binom{r}{0}b^0}^{1\text{ at }r\geq 1}}\right] + \overbrace{0^r}^{1\text{ at }r=0}$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{r}\left[D_{r,b}\binom{x}{b}\right]$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}\sum_{d=0}^{c}\left[D_{c,d}\binom{b}{d}\right]\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}\sum_{d=0}^{0}\left[D_{0,d}\binom{b}{d}\right] + \binom{r}{1}\sum_{d=0}^{1}\left[D_{1,d}\binom{b}{d}\right] + ... + \binom{r}{r-1}\sum_{d=0}^{r-1}\left[D_{r-1,d}\binom{b}{d}\right]\right] + 0^r$$

grouping the terms by $\binom{b}{d}$ tells the coefficents of each diffrence level, this will produce a formula for $D_{r,c}$ in terms of $D_{r,c}$ from lower degree terms.

since every time you lower the degree you loose a diffrence level until D_0, there will be a falling number of each diffrence level in the sum. $\sum^{r-1} D_0 + \sum^{r-2} D_1 + ... \sum^{r-r} D_{r-1}$ (grouping the inner sum, excluding $\sum^{x-1}$)

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}\sum_{d=0}^{0}\left[D_{0,d}\binom{b}{d}\right] + \binom{r}{1}\sum_{d=0}^{1}\left[D_{1,d}\binom{b}{d}\right] + ... + \binom{r}{r-1}\sum_{d=0}^{r-1}\left[D_{r-1,d}\binom{b}{d}\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\overbrace{\left[\binom{r}{0}D_{0,0} + \binom{r}{1}D_{1,0} + ... + \binom{r}{r-1}D_{r-1,0}\right]}^{D_{r,0+1}}\binom{b}{0} + ... + \overbrace{\left[\binom{r}{r-1}D_{r-1,r-1}\right]}^{D_{r,r-1+1}}\binom{b}{r-1}\right] + 0^r$$

beacuse of outer sum the diffrence level has 1 added to it,the grouped coefficents on $\binom{b}{n}$ tells the diffrence level of the nth diffrence as this would expand to n falling sums

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[D_{c,0}\binom{r}{c}\right]\binom{b}{0} + \sum_{c=1}^{r-1}\left[D_{c,1}\binom{r}{c}\right]\binom{b}{1} + \ldots + \sum_{c=r-1}^{r-1}\left[D_{c,r-1}\binom{r}{c}\right]\binom{b}{r-1}\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{b}{c}\sum_{d=c}^{r-1}\left[D_{d,c}\binom{r}{d}\right]\right]\right] + 0^r$$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\ldots\sum_{n_n=0}^{n_{n-1}-1}[1]\right]\right]\right]}_{\text{n sums}} + \binom{0}{n}$$

$$x^r = \sum_{c=0}^{r-1}\left[\overbrace{\binom{x}{c+1}}^{\text{Diffrence sums (c+1)}}\overbrace{\sum_{d=c}^{r-1}\left[D_{d,c}\binom{r}{d}\right]}^{\text{Diffrence coefficents}}\right] + \overbrace{0^r}^{\text{0th Diffrence}}$$

$$D_{r,0} = 0^r$$

$$D_{0,0} = 1$$

$$D_{r,c+1} = \overbrace{\sum_{d=c}^{r-1}\left[D_{d,c}\binom{r}{d}\right]}^{\text{Diffrence coefficents}}$$

$$D_{r,c} = \overbrace{\sum_{d=c-1}^{r-1}\left[D_{d,c-1}\binom{r}{d}\right]}^{\text{Diffrence coefficents}}$$

$$D_{r,c} = \sum_{d=c-1}^{r-1}\left[D_{d,c-1}\binom{r}{d}\right]$$

$$D_{r,c} = \sum_{d=c-1}^{r-1}\left[\sum_{e=c-2}^{d-1}\left[D_{e,c-2}\binom{d}{e}\right]\binom{r}{d}\right]$$

expand c-1 times, (c times if you count the base diffrences identity as the 0th expansion)

$$D_{r,c} = \overbrace{\sum_{d=c-1}^{r-1}\left[\sum_{e=c-2}^{d-1}\left[\sum_{f=c-3}^{e-1}\left[\ldots\sum_{n_{c-1}=c-c}^{n_{c-2}-1}\left[D_{n_{c-1},c-c}\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c\text{ sums}}$$

$$D_{r,c} = \sum_{d=c-1}^{r-1} \left[D_{d,c-1} \binom{r}{d} \right]$$

since x^r only has diffrences upto level r , $D_{r,(c>r)}$ can be treated as 0

$$D_{r,c} = \sum_{d=0}^{r-1} \left[D_{d,c-1} \binom{r}{d} \right]$$

$$D_{r,c} = \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[D_{e,c-2} \binom{d}{e} \right] \binom{r}{d} \right]$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\dots \sum_{n_{c-1}=0}^{n_{c-2}-1} \left[D_{n_{c-1},c-c} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\dots \sum_{n_{c-1}=0}^{n_{c-2}-1} \left[D_{n_{c-1},0} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right]}^{c \text{ sums}}$$

$$D_{r,0} = 0^r$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\dots \sum_{n_{c-1}=0}^{n_{c-2}-1} \left[0^{n_{c-1}} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right]}^{c \text{ sums}}$$

since $0^r = 1$ only at $r = 0$, the inner sums $0^{n_{c-1}}$ will only be 1 at the start, since $0^{\neq 0}$ is 0 the $\binom{n_{c-2}}{n_{c-1}}$ will cancle out when not at the start of the sum, the sum only produces a single non zero run at its start, as long as the inner most upper bound is not negative, if $r \geq c$ the inner sum is guarnted to run at least once

the inner sum only runs once if upper bound is not negative

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}-1}^{n_{c-2}-1}[1]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\sum_{n_{c-2}-1}^{n_{c-2}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c \text{ sums}}$$

for the inner most sum to run the outer sums lower bound must be at least 1, $n_{c-2} \geq 1$, if it is 0 it will not run

$1 - 0^{n_{c-2}}$ is only 1 if $n_{c-2} \neq 0$ or $n_{c-2} > 0$ and $n_{c-2} < 0$

since n_{c-2} starts at 0 and goes upto $n_{c-3} - 1$ it can be negative or its parent sum wouldnt have run it

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[[1-0^{n_{c-2}}]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right)]}^{c-1 \text{ sums}}$$

$$+\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[[-0^{n_{c-2}}]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right)]}^{c-1 \text{ sums}}$$

$$+\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[[-0^{n_{c-2}}]\binom{n_{c-3}}{n_{c-2}}\right]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$-0^{n_{c-2}}$ is only -1 if n_{c-2} is 0, the base of the sum, $\binom{n_{c-3}}{n_{c-2}}$ is 1 at the base

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right)]}^{c-1 \text{ sums}}$$

$$+\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\sum_{n_{c-3}-1}^{n_{c-3}-1}[-1]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$$\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\sum_{n_{c-3}-1}^{n_{c-3}-1}[-1]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}} =$$

$$\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[[-1+0^{n_{c-3}}]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-2 \text{ sums}}$$

$\sum_{b=0}^{a}[0^b]$ will produce 1 if $a \geq 0$, it will produce 0 if $a < 0$ it is a switch that is only on if the upper sum bound is ≥ 0
$\sum_{b=0}^{a}[0^b] = a \geq 0 = a == 0 \mid a > 0$ it defaults to 1 only if $a < 0$ will it be 0, defualt - (1 if less than 0) = 1 - (1 if less than 0)

$$\sum_{b=0}^{a}[0^b] = 1 - \begin{cases} 1 \text{ if } a < 0 \\ 0 \text{ if } a == 0 \\ 0 \text{ if } a > 0 \end{cases} = \begin{cases} 0 \text{ if } a < 0 \\ 1 \text{ if } a == 0 \\ 1 \text{ if } a > 0 \end{cases}$$

$$0^a = \begin{cases} 1 \text{ if } a == 0 \\ 0 \text{ or undefined if } a < 0 \\ 0 \text{ if } a > 0 \end{cases}$$

$$0^{a+1} = \begin{cases} 1 \text{ if } a == (-1) \\ 0 \text{ or undefined if } a < (-1) \\ 0 \text{ if } a > (-1) \end{cases}$$

$1 - 0^{a+1}$ will only be zero if $a = -1$, it is also the inverts 0^{a+1} as it defualts to 1 and is only 0 when $0^{a+1} == 1$

$$1 - 0^{a+1} = \begin{cases} 0 \text{ if } a == (-1) \\ 1 \text{ or undefined if } a < (-1) \\ 1 \text{ if } a > (-1) \end{cases} = \begin{cases} 0 \text{ if } a == (-1) \\ 1 \text{ or undefined if } a < (-1) \\ 1 \text{ if } a == (0) \\ 1 \text{ if } a > (0) \end{cases}$$

2 of the 4 cases match $\begin{cases} 0 \text{ if } a < 0 \\ 1 \text{ if } a == 0 \text{, the 1st case is paritally matched (only for } a == -1 \text{ not for } a < (-1)) \\ 1 \text{ if } a > 0 \end{cases}$

$$\forall a \text{ such that } a == -1 \mid a > 0 \mid a == 0 : \sum_{b=0}^{a}[0^b] = 1 - 0^{a+1}$$

$$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}[0^b] = 1 - 0^a$$

$$\text{let } k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots \sum_{n_n=0}^{n_{n-1}-1}[0_n^n]\ldots\right]\right]\right]}^{n \text{ sums}}$$

$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}[0^b] = 1 - 0^a$, the lower bounds at every sum layer in k have a lower bound of 0, since the upper bound of the next sum is the incrementer-1 and since this incrementer never drops below 0 the identity could be applied to every layer of k. since the identity produces 0^a terms inside a nested sum those terms can have the identity recursivley applied to themselves.The final sum cant be brough out as n_0 is not guranted to to be ≥ 0

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[\sum_{n_{n-1}=0}^{n_{n-2}-1}\left[\sum_{n_n=0}^{n_{n-1}-1}[0_n^n]\right]\right]...\right]\right]\right]}^{n \text{ sums}}$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[\sum_{n_{n-1}=0}^{n_{n-2}-1}[1-0^{n_{n-1}}]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[-(1-0^{n_{n-2}})+\sum_{n_{n-1}=0}^{n_{n-2}-1}[1]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

every expansions inverts the sign of the identity

$$1-0^a = (-1)^0 + (-1)^1 0^a$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[(-1)^1\left((-1)^0+(-1)^1 0^{n_{n-2}}\right)+\sum_{n_{n-1}=0}^{n_{n-2}-1}[(-1)^0]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

LHS and RHS are repeatedly multipled by $(-1)^1$, RHS starts with a degree higher (-1) coefficent

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[(-1)^1+\sum_{n_{n-1}=0}^{n_{n-2}-1}[(-1)^0]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[(-1)^{n-(2)}+(-1)^{n-(2)+1}0^{n_1}+\sum_{n_2=0}^{n_1-1}\left[(-1)^{n-(3)}+\sum_{n_3=0}^{n_2-1}\left[(-1)^{n-(4)}+...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[(-1)^{n-(n-1)}+\sum_{n_{n-1}=0}^{n_{n-2}-1}[(-1)^{n-n}]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

$$k = (-1)^{n-(n)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-1}=0}^{n_{n-2}-1}[1]\right]\right]}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}[1]\right]\right]}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\sum_{n_1=0}^{n_0-1}[(1+(-1)^1 0^{n_1})]}^{1 \text{ sums}}$$

$$= (-1)^{n-(n)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-1}=0}^{n_{n-2}-1}[1]\right]\right]}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}[1]\right]\right]}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\sum_{n_1=0}^{n_0-1}[1]}^{1 \text{ sums}} + (-1)^{n-(2)+1}\overbrace{\sum_{n_1=0}^{n_0-1}[0^{n_1}]}^{1 \text{ sums}}$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(2)+1}\overbrace{\sum_{n_1=0}^{n_0-1}[0^{n_1}]}^{1 \text{ sums}}$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(2)+1}\sum_{n_0-1}^{n_0-1}\left[\underbrace{\overbrace{\binom{n_0}{0}}^{0 \text{ sums}}}_{1} - 0^{n_0}\right]$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(1)}\sum_{n_0-1}^{n_0-1}\left[\underbrace{\overbrace{\binom{n_0}{0}}^{0 \text{ sums}}}_{1}\right] + (-1)^n\sum_{n_0-1}^{n_0-1}[0^{n_0}]$$

$$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}[0^b] = 1-0^a$$

$$\forall n_0 \in \mathbb{Z}, n_0 \geq 0 \Rightarrow$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(1)}\underbrace{\overbrace{\binom{n_0}{0}}^{0 \text{ sums}}}_{1} + (-1)^n 0^{n_0}$$

$$\forall n_0 \in \mathbb{Z}, n_0 \geq 0 \Rightarrow k = \sum_{b=0}^{n-1}\left[(-1)^b\binom{n_0}{n-(b+1)}\right] + (-1)^n 0^{n_0}$$

$$\forall n_0 \in \mathbb{Z}, n_0 \geq 0 \Rightarrow k = \sum_{b=0}^{n-1}\left[(-1)^{n-(b+1)}\binom{n_0}{b}\right] + (-1)^n 0^{n_0}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[0^{n_{c-1}}\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c \text{ sums}}$$

$$\binom{a}{0} = 1$$

only when $b = 0$ does $0^b = 1$, so all other $\binom{a}{b}$ besides the base $\binom{a}{0}$, vanish

$$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}\left[0^b\binom{a}{b}\right] = (1-0^a)\binom{a}{0} = 1 - 0^a$$

this sum is identitcal to k besides each layer of a sum of 1 has binom coefficents

let $n_0 = r$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_c=0}^{n_{c-1}-1}\left[0^{n_c}\binom{n_{c-1}}{n_c}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\sum_{n_c=0}^{n_{c-1}-1}\left[0^{n_c}\binom{n_{c-1}}{n_c}\right]\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[(1-0^{n_{c-1}})\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-1 \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[(-1)^{c-c}\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-1 \text{ sums}} +$$

$$\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[(-1)^{c-(c-1)}\binom{n_{c-3}}{n_{c-2}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-2 \text{ sums}}$$

$$+\ldots+\overbrace{\sum_{n_1=0}^{n_0-1}\left[((-1)^{c-2}+(-1)^{c-2+1}0^{n_1})\binom{n_0}{n_1}\right]}^{1 \text{ sums}}$$

$$D_{r,c} = (-1)^{c-c}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-1 \text{ sums}} +$$

$$(-1)^{c-(c-1)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\binom{n_{c-3}}{n_{c-2}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-2 \text{ sums}}$$

$$+\ldots+(-1)^{c-2}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\binom{n_0}{n_1}\right]}^{1 \text{ sums}} + (-1)^{c-1}\overbrace{\sum_{n_1=0}^{n_0-1}\left[0^{n_1}\binom{n_0}{n_1}\right]}^{1 \text{ sums}}$$

$$\text{binomial therom: } (x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

if both x and $a = 1$ then only the sum of binomial terms will remain as $1^n = 1$

$$(1+1)^r = \sum_{b=0}^{r}\left[\binom{r}{b}(1)(1)\right]$$

$$2^r = \sum_{b=0}^{r}\left[\binom{r}{b}\right]$$

$$2^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}\right] + \overbrace{\binom{r}{r}}^{1}$$

$$2^r - 1 = \sum_{b=0}^{r-1}\left[\binom{r}{b}\right]$$

$$\text{let } l = \sum_{b=0}^{a-1}\left[\binom{a}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}\right]\right] = \sum_{b=0}^{a-1}\left[\binom{a}{b}(2^b-1)\right] = \sum_{b=0}^{a-1}\left[\binom{a}{b}(-1) + \binom{a}{b}(2^b)\right] = (-1)\sum_{b=0}^{a-1}\left[\binom{a}{b}\right] + \sum_{b=0}^{a-1}\left[\binom{a}{b}(2^b)\right]$$

$$\text{let } a' = 1, \text{let } x' = 2$$

$$(1+2)^r = 3^r = \sum_{b=0}^{r}\left[\binom{r}{b}2^b(1)\right]$$

$$3^r - 1(2^r) = \sum_{b=0}^{r-1}\left[\binom{r}{b}2^b(1)\right]$$

$$l = (-1)\sum_{b=0}^{a-1}\left[\binom{a}{b}\right] + 3^a - 2^a = (-1)(2^a-1) + 3^a - 2^a$$

$$(a+b)^r = \sum_{s=0}^{r}\left[\binom{r}{s}a^s b^{r-s}\right]$$

$$\text{let } b = 1$$

$$(a+1)^r = \sum_{s=0}^{r}\left[\binom{r}{s}a^s(1)\right]$$

$$\text{let } a = c+1$$

$$(a+1)^r = \sum_{s=0}^{r}\left[\binom{r}{s}(c+1)^s(1)\right] = \sum_{s=0}^{r}\left[\binom{r}{s}\sum_{s'=0}^{s}\left[\binom{s}{s'}(c)^{s'}\right]\right]$$

this strucutre is almost identical to the diffrences strucutre, the upper bound is the only diffrence, for this to converge to $0^{s''''\cdots}$ the max upper bound must decrease by -1 at each layer, if it does the base of the power term will reach 0. facotr out last sum term to get falling upper bounds.

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}\left[\binom{r}{b}(x-1)^b(1)\right]$$

if x=2 the sum collapses to sum of binomials, if x=1 the sum collapse to a sum of 0^b

apply the identity to each x, here x‘ = x-1

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}[\binom{r}{b}(((x-2)+1)^b(1)]$$

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}(x-2)^c\right]\right]$$

if x=3 the sum collapses to a double sum of binomials, if x=2 the sum collapses to a double sum of 0^c

every expansion increase the number of sums by 1 and decreases the value of the inner x‘ by 1

$$((x-1)+1)^r = x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\right]...\sum_{n_c=0}^{n_{c-1}}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]\right]}^{c\text{ sums}}$$

we can pull out the last term in each sum to get falling sums

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left(\sum_{c=0}^{b-1}\left[\binom{b}{c}\left(...\sum_{n_c=0}^{n_{c-1}-1}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]+\underbrace{\binom{n_{c-1}}{n_{c-1}}}_{1}(x-c)^{n_{c-1}}\right]...\right)+\underbrace{\binom{b}{b}}_{1}(x-2)^b\right)\right]}^{c\text{ sums}}+\underbrace{\binom{r}{r}}_{1}(x-1)^r$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left(\sum_{c=0}^{b-1}\left[\binom{b}{c}\left(...\sum_{n_c=0}^{n_{c-1}-1}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]+(x-c)^{n_{c-1}}\right]...\right)+(x-2)^b\right)\right]}^{c\text{ sums}}+(x-1)^r$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_c=0}^{n_{c-1}-1}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]\right]...\right]}^{c\text{ sums}}$$

$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\binom{n_{c-2}}{n_{c-1}}(x-c)^{n_{c-1}}\right]\right]...\right]}^{c-1\text{ sums}}+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\binom{n_{c-3}}{n_{c-2}}(x-(c-1))^{n_{c-2}}\right]\right]...\right]}^{c-2\text{ sums}}$$

$$+...+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2)^b\right]}^{1\text{ sums}}+(x-1)^r$$